

Arithmetic Progressions in Weierstrass Gap Sets and Garcia–Stichtenoth Towers

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Abstract

For a numerical semigroup S and a positive integer d , let $\text{AP}_d(S)$ be the maximum length of an arithmetic progression of common difference d contained in the gap set of S . For the distinguished Weierstrass semigroups Λ_n in a Garcia–Stichtenoth tower over $\mathbb{F}_{q^2}$, with $q = p^s$, write c_n and g_n for the conductor and genus. We determine the exact p -adic profile $\mathcal{A}_n(t) = \max_{v_p(d)=t} \text{AP}_d(\Lambda_n)$. We prove

$$\mathcal{A}_n(0) = q^{n-1} - 1, \quad \mathcal{A}_n(t) = c_n/p^t \quad (1 \leq t < s),$$

together with the two-branch recursion

$$\mathcal{A}_n(t) = \max \left\{ \left\lceil \frac{c_n - 1}{p^t} \right\rceil, \mathcal{A}_{n-1}(t - s) \right\} \quad (t \geq s).$$

These formulas imply that

$$\text{AP}(\Lambda_n) = \max\{q^{n-1} - 1, c_n/p\}, \quad \frac{g_n}{\text{AP}(\Lambda_n)} \longrightarrow p.$$

A gap AP contributes a triangular family of ordered gap-pair representations, giving a direct lower bound for the Feng–Rao function. For an inductive step, we isolate the contribution of the new parallel gap-AP layer to the classical order bound: its exact discrepancy is twice a predecessor gap-tail count. This gives an AP-layer sharpness criterion and a parity-dependent sharpness window for the tower. More generally, for an inductive sequence $S_k = a_k S_{k-1} \cup [a_k b_k, \infty)$, the standard parameter b_k equals the fixed-difference invariant $\text{AP}_{a_k}(S_k)$, so the known inductive order-bound structure can be decomposed into ancestral AP-layer contributions. This is an AP decomposition of the classical order-bound structure, rather than a new computation of that order bound.

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1 Introduction

Let $S \subseteq \mathbb{N}_0$ be a numerical semigroup and let $G(S) = \mathbb{N}_0 \setminus S$ be its finite gap set. This paper studies the lengths and common differences of arithmetic progressions contained in

$G(S)$ when S occurs geometrically as a Weierstrass semigroup. The central family is the sequence of semigroups at the distinguished rational places of a Garcia–Stichtenoth tower.

The general gap-AP framework used here was introduced in the author’s earlier work [10]. There, for an arbitrary numerical semigroup S , the fixed-difference invariants $\text{AP}_d(S)$, the global invariant $\text{AP}(S)$, and the spectrum $\Sigma(S)$ are studied together with their Apéry- and Kunz-theoretic structure and the associated unrestricted extremal problem. The present article develops a different direction: it specializes the framework to distinguished Weierstrass semigroups in a Garcia–Stichtenoth tower and studies the interaction of their gap-AP structure with Feng–Rao theory.

The questions considered here arise from the interaction of the general gap-AP framework with the recursive geometry of the tower. We determine how large gap arithmetic progressions can be in these asymptotically optimal Weierstrass semigroups and compute their exact p -adic AP profile. We then ask how much Feng–Rao information is carried by the AP spectrum and the parallel conductor-layer progressions. The same analysis leads naturally to inductive numerical semigroups, where the standard blocks admit a direct interpretation through gap-AP invariants and suggest refinements adapted to varying scales.

Additive properties of numerical-semigroup gapsets have already been studied in connection with Buchweitz-type obstructions to being a Weierstrass semigroup, where one considers sumsets such as nG [4]. Our statistic is complementary: it measures additive regularity inside G through long one-dimensional arithmetic progressions. The resulting viewpoint lies naturally between gapset additive combinatorics and the additive-decomposition counts appearing in Feng–Rao theory.

Arithmetic-progressive structure itself is not new in this setting. For example, Silva and Cotterill use residue-class decompositions into arithmetic progressions in their study of weights of $(3, \gamma)$ -hyperelliptic semigroups [15]. Other arithmetic organization of numerical-semigroup gaps has been studied through congruence classes and equidistribution modulo m [14]. These approaches are complementary to ours: congruence statistics describe the distribution of gaps among residue classes, whereas $\text{AP}_d(S)$ records the maximum length of a one-dimensional pattern of fixed difference contained in the gap set. The new point of the present article is the exact p -adic profile of these invariants for the Garcia–Stichtenoth family and the isolation of their conductor-layer contributions inside Feng–Rao counts, rather than the mere occurrence of arithmetic progressions in gap sets or the general definition of $\text{AP}_d(S)$.

Let $q = p^s$, where p is prime, and let c_n and g_n denote the conductor and genus of Λ_n . The tower recursion leads to the p -adic profile

$$\mathcal{A}_n(t) = \max_{\substack{d \geq 1 \\ v_p(d)=t}} \text{AP}_d(\Lambda_n).$$

We prove

$$\begin{aligned} \mathcal{A}_n(0) &= q^{n-1} - 1, \\ \mathcal{A}_n(t) &= c_n/p^t & (1 \leq t < s), \\ \mathcal{A}_n(t) &= \max \left\{ \left\lceil \frac{c_n - 1}{p^t} \right\rceil, \mathcal{A}_{n-1}(t - s) \right\} & (t \geq s). \end{aligned}$$

These formulas give

$$\boxed{\text{AP}(\Lambda_n) = \max \left\{ q^{n-1} - 1, \frac{c_n}{p} \right\}}, \quad \boxed{\lim_{n \rightarrow \infty} \frac{g_n}{\text{AP}(\Lambda_n)} = p.}$$

The asymptotic genus-to-AP ratio therefore detects the characteristic rather than the tower parameter.

The profile also has a coding-theoretic interpretation. A k -term gap AP contributes a triangular family of ordered gap-pair representations, producing a direct AP-to-Feng–Rao inequality. At an inductive step $S = aT \cup [C, \infty)$, the new conductor layer consists of parallel gap APs. Writing δ_i for the indexed order-bound sequence, their contribution gives a natural lower estimate L_i . For an explicit predecessor threshold m_i , the principal coding-theoretic decomposition is

$$\delta_i - L_i = 2\#\{h \in G(T) : h > m_i\}.$$

It separates the part determined solely by the current parallel AP layer from the predecessor-tail correction in the full classical order bound. The associated sharpness theorem identifies precisely when this AP-layer estimate, rather than merely the order-bound sequence itself, has stabilized to equality. For the Garcia–Stichtenoth tower the sharpness window has length q at odd levels and formal parameter $q^{n/2}(q-1) + q$ at even levels.

At common difference one, Garcia, Kim, and Lax used blocks of consecutive Weierstrass gaps to improve designed-distance estimates [7]. Their argument depends on the intermediate integers in a full gap interval, so a d -spaced progression does not give a formal extension. Our approach instead uses the standard gap-pair expression and recovers, from the gap side, the tower recursion of Bras-Amorós and O’Sullivan [2].

The same layer picture extends to inductive semigroups, whose block structure, genus formulas, Apéry-set recursions, and Feng–Rao invariants are established [3, 5]. For a sequence $S_k = a_k S_{k-1} \cup [a_k b_k, \infty)$, we read the standard blocks on the complementary gap side as AP layers and prove the new identification

$$b_k = \text{AP}_{a_k}(S_k).$$

This converts the known inductive genus formula into the AP interpretation

$$g(S_n) = \sum_{k=2}^n (a_k - 1) \text{AP}_{a_k}(S_k)$$

and expresses the predecessor-tail discrepancy as a sum of ancestral gap-AP contributions. For the tower, a q -depth AP profile recovers these ancestral gap-AP lengths from a single level.

In this sense, the earlier work [10] develops the unrestricted numerical-semigroup theory of gap arithmetic progressions, whereas the present article develops their behavior in a distinguished geometric family and their interaction with one-point coding theory.

The argument therefore passes from maximal fixed-difference gap APs to the exact Garcia–Stichtenoth p -adic profile, then to convolution in gap-pair counts, and finally to the decomposition of the Feng–Rao contribution into the current AP layer and its predecessor and ancestral corrections.

The Garcia–Stichtenoth tower is asymptotically optimal with respect to the Drinfeld–Vlăduț bound [18, 8, 9]. Such towers supply geometric input for asymptotic algebraic-geometry code constructions in the circle of the Tsfasman–Vlăduț–Zink bound [17]. We do not say that a numerical semigroup itself achieves a coding bound.

The paper is organized as follows. Section 2 recalls the AP invariants, and Section 3 recalls the required Weierstrass-semigroup and coding background. Sections 4–6 establish the Garcia–Stichtenoth formulas, the p -adic AP profile, and its asymptotic consequences. Section 7 develops the gap-pair convolution and the AP-layer decomposition of the tower

order bound. Section 8 reinterprets known inductive block and genus structure through gap-AP invariants and introduces the tower’s q -depth profile. Section 9 records the remaining problems.

Remark 1.1 (Novelty status). The fixed-difference gap-AP invariants used here were introduced in [10]. The recursive structure of the Garcia–Stichtenoth semigroups, the classical Feng–Rao order bound, the theory of Arf and acute semigroups, and the standard block and genus formulas for inductive numerical semigroups are established [1, 2, 3, 5]. Arithmetic-progressive, sumset, and congruence structures in special gap sets have also been studied previously [15, 4, 14]. The contributions of the present article are the exact p -adic AP profile and global AP value for the Garcia–Stichtenoth family, the characteristic slope, and the AP-layer decomposition and sharpness analysis of Feng–Rao information. We do not claim novelty for the underlying inductive structure or the classical order bound itself.

2 Arithmetic progressions in numerical-semigroup gaps

Definition 2.1. A *numerical semigroup* is a submonoid $S \subseteq \mathbb{N}_0$ with finite complement. Its gap set, genus, conductor, and multiplicity are respectively

$$G(S) = \mathbb{N}_0 \setminus S, \quad g(S) = |G(S)|, \quad c(S) = \min\{c : c + \mathbb{N}_0 \subseteq S\},$$

and

$$m(S) = \min(S \setminus \{0\}).$$

We use standard numerical-semigroup terminology as in [13].

Definition 2.2. Following [10], for $d \geq 1$ we write

$$\text{AP}_d(S) = \max\{k : \text{there is } x \geq 1 \text{ with } x, x + d, \dots, x + (k - 1)d \in G(S)\}.$$

If $G(S) = \emptyset$, we set $\text{AP}_d(S) = 0$. We define

$$\text{AP}(S) = \max_{d \geq 1} \text{AP}_d(S)$$

and the AP spectrum

$$\Sigma(S) = \{(d, k) \in \mathbb{N}^2 : \text{AP}_d(S) \geq k\}.$$

Lemma 2.3. For every numerical semigroup S ,

$$\text{AP}_1(S) = m(S) - 1.$$

Proof. The integers $1, \dots, m(S) - 1$ are gaps. Any block of $m(S)$ consecutive positive integers contains a positive multiple of $m(S)$ and is therefore not entirely contained in the gap set. $\square$

The scalar invariant $\text{AP}(S)$ forgets most of the spectrum. For a recursion built from multiplication by $q = p^s$, common differences with distinct p -adic valuations behave differently. The earlier work analyzes the general distinction between semigroup differences and gap differences through Apéry sets and Kunz coordinates [10]. In the tower considered here, the scaled recursion leads to the profile in Definition 5.1 below.

3 Weierstrass semigroups and one-point codes

Let X be a smooth projective curve of genus g over a field k , and let P be a rational point. Its Weierstrass semigroup is

$$H(P) = \{n \geq 0 : \text{there is } f \in k(X) \text{ whose only pole is at } P \text{ and has order } n\}.$$

It is a numerical semigroup. The Weierstrass gap theorem gives

$$|G(H(P))| = g$$

for every point P ; this statement is not restricted to non-Weierstrass points. A non-Weierstrass point on a classical curve has the generic gap sequence $1, 2, \dots, g$. See [16] for the function-field formulation.

For minimal coding background, let $D = P_1 + \dots + P_N$ be a sum of rational points disjoint from P . Evaluation of functions in $\mathcal{L}(mP)$ at the P_i gives a one-point AG code; equivalent dual conventions are common in the order-bound literature. General background is available in [11, 16].

We use exactly the indexing convention of Bras-Amorós and O’Sullivan [2]. If

$$H(P) = \{\rho_0 = 0 < \rho_1 < \rho_2 < \dots\},$$

then

$$\nu_i = \#\{j \in \mathbb{N}_0 : \rho_i - \rho_j \in H(P)\} = \#\{(j, k) \in \mathbb{N}_0^2 : \rho_j + \rho_k = \rho_i\}, \quad (1)$$

and the corresponding indexed order-bound sequence is

$$\delta_i = \min\{\nu_j : j > i\}. \quad (2)$$

Other sources shift i by one; formulas should not be transferred between conventions without checking that shift. The original decoding framework is due to Feng and Rao [6].

Conceptually, ν_i counts additive decompositions among nongaps, whereas $\Sigma(H(P))$ measures additive regularity in the gaps. Complementarity alone does not produce a useful inequality, but it makes a recursive comparison natural.

There is also a classical precedent at common difference one. Garcia, Kim, and Lax showed that suitable blocks of consecutive Weierstrass gaps can improve the usual designed-distance estimate for certain Goppa codes [7]. In the present notation this is the $d = 1$ layer. Their interval argument uses the intermediate integers in a full gap block. For $d > 1$, an arithmetic progression leaves those intermediate integers uncontrolled. The formal substitution

$$\text{consecutive gaps} \quad \longmapsto \quad d\text{-spaced gaps}$$

is therefore not justified. The argument below instead begins with the gap-pair expression for the Feng–Rao function.

4 The Garcia–Stichtenoth Weierstrass semigroups

Fix a prime power $q = p^s$. We use the indexing of [2], which in turn cites [9, 12]. The tower is defined over $\mathbb{F}_{q^2}$ by

$$\mathcal{F}^1 = \mathbb{F}_{q^2}(x_1), \quad \mathcal{F}^n = \mathcal{F}^{n-1}(x_n),$$

where

$$x_n^q + x_n = \frac{x_{n-1}^q}{x_{n-1}^{q-1} + 1}. \quad (3)$$

Let Q^n be the rational place that is the unique pole of x_1 in $\mathcal{F}^n$. The Weierstrass semigroup $\Lambda_n = H(Q^n)$ satisfies

$$\Lambda_1 = \mathbb{N}_0, \quad \Lambda_n = q\Lambda_{n-1} \cup [c_n, \infty) \quad (n \geq 2), \quad (4)$$

where

$$c_n = q^n - q^{\lfloor (n+1)/2 \rfloor}. \quad (5)$$

This is the conductor normalization used throughout the paper.

The genus is given by

$$g_n = \left(q^{\lfloor (n+1)/2 \rfloor} - 1 \right) \left(q^{\lceil (n-1)/2 \rceil} - 1 \right). \quad (6)$$

Equivalently, its even and odd values are

$$g_{2r} = (q^r - 1)^2, \quad g_{2r+1} = (q^{r+1} - 1)(q^r - 1),$$

and $g_n \sim q^n$ for fixed q .

Below c_n , recursion (4) has the useful complement form

$$x \in G(\Lambda_n) \iff x < c_n \text{ and } (q \nmid x \text{ or } x/q \in G(\Lambda_{n-1})). \quad (7)$$

We also use

$$qc_{n-1} \leq c_n. \quad (8)$$

Equality holds for odd n ; the inequality is strict for even n .

Proposition 4.1. *For $n \geq 2$,*

$$m(\Lambda_n) = q^{n-1}.$$

It follows in particular that

$$\text{AP}(\Lambda_n) \geq \text{AP}_1(\Lambda_n) = q^{n-1} - 1.$$

Proof. Inductively, the least positive element of $q\Lambda_{n-1}$ is q^{n-1} . It remains to check that the conductor layer does not introduce a smaller positive element. With $r = \lfloor (n+1)/2 \rfloor$, we have

$$c_n - q^{n-1} = q^{n-1}(q-1) - q^r \geq 0,$$

because $n-1 \geq r$ and $q^{n-1-r}(q-1) \geq 1$. It follows that the least positive nongap is q^{n-1} . The AP statement follows from Lemma 2.3. $\square$

5 The p -adic AP profile

Definition 5.1. For $n \geq 1$ and $t \geq 0$, define

$$\mathcal{A}_n(t) = \max_{\substack{d \geq 1 \\ v_p(d)=t}} \text{AP}_d(\Lambda_n).$$

For $n = 1$ the profile is identically zero. We call $\mathcal{A}_n$ the p -adic AP profile.

The layer in which the common difference is coprime to p is controlled by the multiplicity computed in Proposition 4.1.

Theorem 5.2 (Exact p -coprime profile). *For every $n \geq 2$,*

$$\boxed{\mathcal{A}_n(0) = q^{n-1} - 1.}$$

Proof. Let d satisfy $p \nmid d$. Since $q = p^s$,

$$\gcd(d, q^{n-1}) = 1.$$

Any q^{n-1} consecutive terms of a d -progression therefore run through all residue classes modulo q^{n-1} . One term is a positive multiple of $q^{n-1} = m(\Lambda_n)$ and hence is a nongap. We conclude that $\text{AP}_d(\Lambda_n) \leq q^{n-1} - 1$. Equality is attained for $d = 1$ by

$$1, 2, \dots, q^{n-1} - 1.$$

□

When the valuation lies strictly between 0 and s , the conductor gives a sharp bound without requiring a separate count of the periodically q -divisible terms.

Theorem 5.3 (Intermediate p -adic profile). *For $n \geq 2$ and $1 \leq t < s$,*

$$\boxed{\mathcal{A}_n(t) = \frac{c_n}{p^t}.$$

Proof. First note that $q \mid c_n$, so $p^t \mid c_n$. The progression

$$1, 1 + p^t, 1 + 2p^t, \dots, 1 + \left(\frac{c_n}{p^t} - 1\right)p^t \quad (9)$$

has c_n/p^t terms and last term $c_n - p^t + 1 < c_n$. Every term is congruent to 1 modulo p , so none is divisible by q . By (7), all of them are gaps. This proves the lower bound.

Conversely, suppose $x, x + d, \dots, x + (k - 1)d$ consists of gaps and $v_p(d) = t$. Every gap is less than c_n , while $x \geq 1$ and $d \geq p^t$. Therefore

$$x + (k - 1)d \leq c_n - 1 \implies k \leq \left\lceil \frac{c_n - 1}{p^t} \right\rceil = \frac{c_n}{p^t}.$$

The last equality uses $p^t \mid c_n$. This proves the upper bound. □

Remark 5.4. The case split according to $p^t \mid x$ is valid but unnecessary for this theorem. The universal conductor bound already matches the explicit progression (9).

For differences divisible by q , the progression remains in a single residue class modulo q . The following fixed-difference statement separates the nonzero residue classes from the divisible class and supplies the higher-valuation recursion.

Proposition 5.5 (Fixed-difference recursion). *Let $n \geq 2$ and let d be divisible by q . One has*

$$\boxed{\text{AP}_d(\Lambda_n) = \max \left\{ \left\lceil \frac{c_n - 1}{d} \right\rceil, \text{AP}_{d/q}(\Lambda_{n-1}) \right\}.$$

Proof. Because $q \mid d$, all terms of a d -progression have the same residue modulo q .

If $q \nmid x$, no term is divisible by q . Such terms are gaps exactly as long as they remain below c_n . The maximum length in this branch is

$$\left\lceil \frac{c_n - 1}{d} \right\rceil,$$

attained by $x = 1$.

If $q \mid x$, every term is divisible by q . Dividing the progression by q gives common difference d/q . By (7), a divisible integer below c_n is a gap of Λ_n precisely when its quotient is a gap of Λ_{n-1} . Conversely, scaling a gap progression in Λ_{n-1} by q gives gaps of Λ_n : all scaled terms lie below $qc_{n-1} \leq c_n$. It follows that the best length in this branch is $\text{AP}_{d/q}(\Lambda_{n-1})$.

These two cases exhaust all possible initial residues modulo q . $\square$

Theorem 5.6 (Recursive profile). *For $n \geq 2$ and $t \geq s$,*

$$\mathcal{A}_n(t) = \max \left\{ \left\lceil \frac{c_n - 1}{p^t} \right\rceil, \mathcal{A}_{n-1}(t - s) \right\}.$$

Proof. Take the maximum in Proposition 5.5 over all d with $v_p(d) = t$. The first term is largest for the smallest such difference, $d = p^t$. The map $d \mapsto d/q$ is a bijection from positive integers of p -adic valuation t to those of valuation $t - s$, so the second term maximizes to $\mathcal{A}_{n-1}(t - s)$. $\square$

Remark 5.7 (Endpoint audit). When $v_p(d) > s$, an initial term x may have any valuation between 0 and $s - 1$. This creates no third branch: since $q \mid d$, such an x remains nonzero modulo q throughout the progression and belongs to the first branch of Proposition 5.5.

6 Exact AP length and asymptotics

Theorem 6.1 (Exact global AP length). *For $n \geq 2$,*

$$\text{AP}(\Lambda_n) = \max \left\{ q^{n-1} - 1, \frac{c_n}{p} \right\}.$$

Equivalently,

$$\text{AP}(\Lambda_n) = \begin{cases} q^{n-1} - 1, & q = p, \\ \frac{q^n - q^{\lfloor (n+1)/2 \rfloor}}{p}, & q = p^s, \ s \geq 2. \end{cases}$$

Proof. Let $B_n = \max\{q^{n-1} - 1, c_n/p\}$. Theorem 5.2 gives the first lower bound. The progression $1, 1 + p, \dots$ below c_n gives the second; for $s = 1$ it is the nonzero residue branch in Proposition 5.5, and for $s \geq 2$ it is Theorem 5.3 with $t = 1$.

It remains to bound every profile layer by B_n . The layers $t = 0$ and $1 \leq t < s$ are handled by Theorems 5.2 and 5.3; among the latter, $c_n/p^t \leq c_n/p$. For $t \geq s$, Theorem 5.6 gives

$$\mathcal{A}_n(t) \leq \max\{c_n/q, \text{AP}(\Lambda_{n-1})\}.$$

Now

$$\frac{c_n}{q} = q^{n-1} - q^{\lfloor (n+1)/2 \rfloor - 1} \leq q^{n-1} - 1 \leq B_n.$$

Inductively, $\text{AP}(\Lambda_{n-1}) = B_{n-1} \leq B_n$, since both $q^{n-2} - 1 \leq q^{n-1} - 1$ and $c_{n-1}/p \leq c_n/p$. This proves the first boxed formula.

If $q = p$, then

$$c_n/p = q^{n-1} - q^{\lfloor (n+1)/2 \rfloor - 1} \leq q^{n-1} - 1.$$

If $s \geq 2$, then $q \geq p^2$, so $q - 1 \geq p$, and

$$c_n = q^{n-1}(q - 1) + (q^{n-1} - q^{\lfloor (n+1)/2 \rfloor}) \geq pq^{n-1} > p(q^{n-1} - 1).$$

It follows that c_n/p is the larger term. This proves the piecewise form, including the case $n = 2$. $\square$

The proof also identifies explicit maximizers. If $q = p$ is prime, the progression

$$1, 2, \dots, q^{n-1} - 1$$

is maximizing. If $q = p^s$ with $s \geq 2$, a maximizing progression is

$$1, 1 + p, 1 + 2p, \dots, 1 + \left(\frac{c_n}{p} - 1\right)p.$$

Its last term is exactly $c_n - p + 1 < c_n$, and every term is nonzero modulo p and therefore does not belong to $q\Lambda_{n-1}$.

Corollary 6.2 (Characteristic slope). *For fixed $q = p^s$,*

$$\boxed{\lim_{n \rightarrow \infty} \frac{g_n}{\text{AP}(\Lambda_n)} = p.}$$

Proof. Formula (6) gives $g_n \sim q^n$. If $s = 1$, then $\text{AP}(\Lambda_n) = q^{n-1} - 1 \sim q^n/p$. If $s \geq 2$, then $\text{AP}(\Lambda_n) = c_n/p \sim q^n/p$. The quotient therefore tends to p . $\square$

7 Gap arithmetic progressions and the Feng–Rao function

The passage from gap arithmetic progressions to the Feng–Rao function begins with the standard gap-pair identity. The Bras–Amorós–O’Sullivan recursion for the Garcia–Stichtenoth tower is also known; the contribution developed here is the AP convolution and its interpretation through parallel gap layers. For a numerical semigroup S , denote its gap set, genus, and conductor by

$$G = G(S), \quad g = |G|, \quad c = c(S).$$

For an integer λ , let

$$R_S(\lambda) = \#\{h \in G : \lambda - h \in G\} = \#\{(h_1, h_2) \in G^2 : h_1 + h_2 = \lambda\}.$$

The quantity R_S counts ordered gap pairs. We use the value notation

$$\nu_S(\lambda) = \#\{s \in S : \lambda - s \in S\}.$$

When $S = \{\rho_0 = 0 < \rho_1 < \dots\}$, the convention of (1) is $\nu_i = \nu_S(\rho_i)$.

Lemma 7.1 (Standard gap-pair identity). *For every $\lambda \geq c$,*

$$\boxed{\nu_S(\lambda) = \lambda + 1 - 2g + R_S(\lambda).}$$

Proof. This is the standard gap-pair form of the Feng–Rao divisor count; see, for example, the numerical-semigroup treatment in [11]. A direct verification uses the indicator functions $\mathbf{1}_S$ and $\mathbf{1}_G$ on $\mathbb{N}_0$. Since S and G partition $\mathbb{N}_0$,

$$\mathbf{1}_S(n) = 1 - \mathbf{1}_G(n).$$

All gaps lie in $[0, \lambda]$ when $\lambda \geq c$, and hence

$$\begin{aligned} \nu_S(\lambda) &= \sum_{a=0}^{\lambda} \mathbf{1}_S(a) \mathbf{1}_S(\lambda - a) \\ &= \lambda + 1 - \sum_{a=0}^{\lambda} \mathbf{1}_G(a) - \sum_{a=0}^{\lambda} \mathbf{1}_G(\lambda - a) + \sum_{a=0}^{\lambda} \mathbf{1}_G(a) \mathbf{1}_G(\lambda - a) \\ &= \lambda + 1 - 2g + R_S(\lambda). \end{aligned}$$

□

The identity therefore converts lower bounds and recursions for gap-pair counts into corresponding statements for the Feng–Rao function.

An arithmetic progression inside the gap set supplies such a lower bound through its additive convolution. Let

$$P = \{x, x + d, \dots, x + (k - 1)d\} \subseteq G.$$

For each $0 \leq r \leq 2k - 2$, define

$$\lambda_r = 2x + rd$$

and let the associated triangular function be

$$\tau_k(r) = \#\{(i, j) \in \{0, \dots, k - 1\}^2 : i + j = r\} = \begin{cases} r + 1, & 0 \leq r \leq k - 1, \\ 2k - 1 - r, & k - 1 \leq r \leq 2k - 2. \end{cases}$$

Proposition 7.2 (Gap-AP convolution bound). *For $0 \leq r \leq 2k - 2$,*

$$\boxed{R_S(\lambda_r) \geq \tau_k(r).}$$

Consequently, whenever $\lambda_r \geq c$,

$$\boxed{\nu_S(\lambda_r) \geq \lambda_r + 1 - 2g + \tau_k(r).}$$

Proof. Every ordered pair (i, j) with $i + j = r$ gives the ordered gap-pair representation

$$(x + id) + (x + jd) = 2x + rd = \lambda_r.$$

Distinct ordered index pairs give distinct ordered pairs of gaps, proving the first inequality. The second follows from Lemma 7.1. □

This is a direct AP-to-gap-pair, and hence AP-to-Feng–Rao, inequality. Although elementary once the progression is fixed, this formulation does not appear explicitly in the literature known to the author. We use it here without making a universal priority claim.

No sharpness assertion is made: other gaps may give additional representations of the same target.

At the central coefficient $r = k - 1$, the proposition gives the particularly simple estimate

$$R_S(2x + (k - 1)d) \geq k.$$

To make the convolution exact rather than merely bounded below, consider an inductive step

$$S = aT \cup [C, \infty), \quad a \geq 2, \quad C = ab, \quad C \geq ac(T). \quad (10)$$

Below the conductor, the gap set decomposes as

$$\boxed{G(S) = U_{a,C} \sqcup aG(T)}, \quad U_{a,C} = \{1 \leq x < C : a \nmid x\}. \quad (11)$$

The new layer further decomposes as

$$U_{a,C} = \bigsqcup_{r=1}^{a-1} P_r, \quad P_r = \{r, r + a, \dots, r + (b - 1)a\}. \quad (12)$$

Consequently, the new conductor layer of gaps is a union of $a - 1$ parallel arithmetic progressions, all of common difference a and length $b = C/a$.

For $C \leq \lambda < 2C$, let $\Delta = 2C - \lambda$ and define

$$B_{a,C}(\lambda) = \#\{(u, v) \in U_{a,C}^2 : u + v = \lambda\}.$$

The convolution of the parallel layer is

$$\boxed{B_{a,C}(\lambda) = \begin{cases} \Delta - \frac{\Delta}{a}, & a \mid \Delta, \\ \Delta - 1 - 2 \left\lfloor \frac{\Delta}{a} \right\rfloor, & a \nmid \Delta. \end{cases}} \quad (13)$$

To compute this convolution, make the change of variables

$$u = C - y, \quad v = C - (\Delta - y).$$

The conditions $1 \leq u, v < C$ are equivalent to $1 \leq y \leq \Delta - 1$. Because $a \mid C$, the two entries belong to $U_{a,C}$ exactly when neither y nor $\Delta - y$ is divisible by a . If $a \mid \Delta$, these forbidden conditions coincide and exclude $\Delta/a - 1$ values. If $a \nmid \Delta$, they are distinct residue classes and each excludes $\lfloor \Delta/a \rfloor$ values. Subtracting from $\Delta - 1$ proves (13), including the endpoints $\Delta = 1$ and $\Delta = C$.

The decomposition of the gap set also separates the inherited contribution, which is measured by

$$N_T(\lambda) = \#\{h \in G(T) : 0 < \lambda - ah < C\}.$$

Theorem 7.3 (Gap-pair recursion for an inductive step). *Under (10), for $C \leq \lambda < 2C$,*

$$\boxed{R_S(\lambda) = B_{a,C}(\lambda) + \begin{cases} R_T(\lambda/a), & a \mid \lambda, \\ 2N_T(\lambda), & a \nmid \lambda. \end{cases}}$$

Proof. Use the disjoint decomposition (11). Pairs in $U_{a,C}^2$ contribute $B_{a,C}(\lambda)$.

If $a \mid \lambda$, a mixed pair consisting of one element of $U_{a,C}$ and one element of $aG(T)$ cannot have sum divisible by a . The remaining pairs are two scaled gaps, and division by a identifies them with the ordered gap pairs of T summing to λ/a .

If $a \nmid \lambda$, two scaled gaps cannot sum to λ . A gap $ah \in aG(T)$ has a unique partner $u = \lambda - ah$ in $U_{a,C}$ precisely when $0 < u < C$; nondivisibility of u is automatic. The scaled gap can occur in either coordinate, giving the factor 2. $\square$

Combining this theorem with the standard gap-pair identity gives the known shape of the Feng–Rao recursion.

Corollary 7.4 (Feng–Rao recursion in value notation). *Under (10), for $C \leq \lambda < 2C$,*

$$\nu_S(\lambda) = \begin{cases} \nu_T(\lambda/a), & a \mid \lambda, \\ 2\#\{t \in T : at < \lambda - C\}, & a \nmid \lambda. \end{cases}$$

The inequality in the second branch is strict, and $0 \in T$ contributes. The endpoint $\lambda = C$ belongs to the divisible branch.

Proof. The gap decomposition gives

$$g(S) = (a - 1)b + g(T).$$

If $a \mid \lambda$, then $a \mid \Delta$, and Theorem 7.3 and (13) give

$$\begin{aligned} \nu_S(\lambda) &= \lambda + 1 - 2(a - 1)b + R_T(\lambda/a) + \Delta - \Delta/a - 2g(T) \\ &= \lambda/a + 1 - 2g(T) + R_T(\lambda/a) = \nu_T(\lambda/a). \end{aligned}$$

Here $\lambda/a \geq b \geq c(T)$, so the gap-pair identity applies to T .

Suppose $a \nmid \lambda$. Express the difference from the conductor as

$$\lambda - C = am + r, \quad 0 \leq m \leq b - 1, \quad 1 \leq r \leq a - 1.$$

The relation above gives $\lfloor \Delta/a \rfloor = b - m - 1$. The mixed condition defining N_T reduces to $h \geq m + 1$: its upper inequality is automatic because every gap of T is less than $c(T) \leq b$. Consequently,

$$N_T(\lambda) = g(T) - \#\{h \in G(T) : h \leq m\}.$$

Substitution in Lemma 7.1, using (13), yields

$$\begin{aligned} \nu_S(\lambda) &= 2(m + 1 - g(T) + N_T(\lambda)) \\ &= 2\#\{t \in T : 0 \leq t \leq m\} = 2\#\{t \in T : at < \lambda - C\}. \end{aligned}$$

□

The same inductive decomposition now visibly controls both invariants. For gap arithmetic progressions, a difference divisible by a either stays in a nonzero residue class below the conductor or descends after division by a . For the Feng–Rao function, a target value divisible by a descends to the predecessor semigroup, whereas a nondivisible target is controlled by the new conductor layer. The AP recursion and the Feng–Rao recursion are therefore two manifestations of the same scaled-semigroup decomposition.

The Garcia–Stichtenoth recursion is obtained by applying Theorem 7.3 with

$$S = \Lambda_n, \quad T = \Lambda_{n-1}, \quad a = q, \quad C = c_n.$$

The new gap layer is

$$U_n = \{1 \leq x < c_n : q \nmid x\} = \bigsqcup_{r=1}^{q-1} \left\{ r, r + q, \dots, r + \left(\frac{c_n}{q} - 1 \right) q \right\}.$$

It consists of $q - 1$ parallel APs of length c_n/q . Write $R_n = R_{\Lambda_n}$. For $c_n \leq \lambda < 2c_n$, let $\Delta = 2c_n - \lambda$. The preceding formulas specialize to

$$B_n(\lambda) = \begin{cases} \Delta - \frac{\Delta}{q}, & q \mid \Delta, \\ \Delta - 1 - 2 \left\lfloor \frac{\Delta}{q} \right\rfloor, & q \nmid \Delta, \end{cases} \quad (14)$$

and

$$R_n(\lambda) = B_n(\lambda) + \begin{cases} R_{n-1}(\lambda/q), & q \mid \lambda, \\ 2N_{n-1}(\lambda), & q \nmid \lambda, \end{cases} \quad (15)$$

where

$$N_{n-1}(\lambda) = \#\{h \in G(\Lambda_{n-1}) : 0 < \lambda - qh < c_n\}.$$

The gap-pair identity consequently gives

$$\nu_{\Lambda_n}(\lambda) = \begin{cases} \nu_{\Lambda_{n-1}}(\lambda/q), & q \mid \lambda, \\ 2\#\{t \in \Lambda_{n-1} : qt < \lambda - c_n\}, & q \nmid \lambda. \end{cases} \quad (16)$$

After translating $\lambda = \rho_i = i + g_n$ above the conductor, the range $c_n < \lambda < 2c_n$ in (16) is exactly the divisible/nondivisible transition recursion proved by Bras-Amorós and O’Sullivan [2, Lemma 5]; the endpoint $\lambda = c_n$ is compatible with their preceding branch. In the present AP formulation, the known recursion is obtained from the convolution of the parallel gap layer.

The calculation is illustrated by the following rows for $q = 3$ and $n = 3$, where $c_3 = 18$. The column E is the inherited contribution when $3 \mid \lambda$ and the mixed contribution otherwise.

λ	$B_3(\lambda)$	E	$R_3(\lambda)$	$\nu_{\Lambda_3}(\lambda)$	branch
27	6	2	8	4	inherited
28	3	4	7	4	mixed
29	2	4	6	4	mixed
30	4	1	5	4	inherited
31	2	2	4	4	mixed
32	1	2	3	4	mixed

In every row $R_3(\lambda) = B_3(\lambda) + E$; the table is illustrative, not part of the proof.

The exact recursion contains a universal contribution from the new parallel layer and a nonnegative inherited or mixed term. Dropping the latter in Theorem 7.3 gives an immediate coding-theoretic consequence of the parallel AP layer alone.

Corollary 7.5 (AP-layer bound for the Feng–Rao function). *Under (10), for $C \leq \lambda < 2C$,*

$$\boxed{\nu_S(\lambda) \geq \lambda + 1 - 2g(S) + B_{a,C}(\lambda).}$$

To pass from the Feng–Rao function to the order-bound sequence, we minimize this estimate over the later nongaps. For $\Delta \geq 1$, define

$$\varepsilon_a(\Delta) = \begin{cases} \Delta - 1 - 2 \left\lfloor \frac{\Delta}{a} \right\rfloor, & a \nmid \Delta, \\ \Delta + 1 - \frac{2\Delta}{a}, & a \mid \Delta. \end{cases}$$

Corollary 7.6 (AP-derived order-bound estimate). *Let $S = \{\rho_0 = 0 < \rho_1 < \dots\}$ satisfy (10), and let $\delta_i = \min_{j>i} \nu_S(\rho_j)$. If*

$$C \leq \rho_{i+1} < 2C,$$

equivalently $C - g(S) - 1 \leq i \leq 2C - g(S) - 2$, then

$$\rho_{i+1} = i + 1 + g(S), \quad \Delta_i = 2C - \rho_{i+1},$$

and

$$\boxed{\delta_i \geq i - g(S) + 2 + \varepsilon_a(\Delta_i).}$$

Proof. Let $\lambda_0 = \rho_{i+1}$ and $\Delta_i = 2C - \lambda_0$. Since every integer at least C is a nongap, the later targets are all integers $\lambda \geq \lambda_0$. For $\lambda < 2C$, let $D = 2C - \lambda$. Corollary 7.5 gives

$$\nu_S(\lambda) \geq \lambda_0 + 1 - 2g(S) + \Delta_i - D + B_{a,C}(2C - D).$$

For $\lambda \geq 2C$, the standard gap-pair identity and nonnegativity of R_S give $\nu_S(\lambda) \geq \lambda + 1 - 2g(S)$, whose smallest relevant value is obtained at $2C$. Consequently, the correction to the baseline $\lambda_0 + 1 - 2g(S) = i - g(S) + 2$ is at least

$$\min_{0 \leq D \leq \Delta_i} \{\Delta_i - D + B_{a,C}(2C - D)\}, \quad (17)$$

where the term at $D = 0$ is interpreted as Δ_i .

If D is a positive multiple of a , then $B_{a,C}(2C - D) - D = -D/a$. Otherwise it equals $-1 - 2 \lfloor D/a \rfloor$. If $a \nmid \Delta_i$, the minimum is attained at $D = \Delta_i$ and equals $-1 - 2 \lfloor \Delta_i/a \rfloor$. If $a \mid \Delta_i$, the largest nonmultiple $D = \Delta_i - 1$ gives $1 - 2\Delta_i/a$, which is no larger than the value $-\Delta_i/a$ at the largest multiple; equality occurs when $\Delta_i = a$. Substitution in (17) gives exactly $\varepsilon_a(\Delta_i)$. $\square$

For $a \geq 3$, the correction $\varepsilon_a(\Delta)$ is positive for every $\Delta \geq 2$ and is zero for $\Delta = 1$. It therefore gives a positive correction over the baseline linear term throughout $C \leq \rho_{i+1} \leq 2C - 2$. For $a = 2$, the correction is 1 for even Δ and 0 for odd Δ . This gives an AP-derived lower estimate for the order-bound sequence, whose discrepancy from the exact value is controlled by the predecessor gaps.

Inductive numerical semigroups lie in the Arf setting, and the classical Feng–Rao distance for Arf semigroups has been computed explicitly in important ranges [3, 5]. More generally, Bras-Amorós introduced acute semigroups, a class containing Arf semigroups, and determined a stabilization threshold beyond which the order bound is the next ν -value [1]. Here we instead isolate the part arising from the parallel arithmetic-progression layer in the gap set and compute its exact discrepancy from the full order bound.

For an index in the range of Corollary 7.6, define

$$k_i = \left\lfloor \frac{\Delta_i - 1}{a} \right\rfloor, \quad m_i = b - 1 - k_i = b - 1 - \left\lfloor \frac{\Delta_i - 1}{a} \right\rfloor. \quad (18)$$

Theorem 7.7 (Predecessor form of the transition-range order bound). *Let $S = aT \cup [C, \infty)$ satisfy (10), so $C = ab$ and $b \geq c(T)$. For every index satisfying $C \leq \rho_{i+1} < 2C$,*

$$\boxed{\delta_i = 2\#\{t \in T : 0 \leq t \leq m_i\}.}$$

Equivalently,

$$\boxed{\delta_i = 2\#\left\{t \in T : 0 \leq t \leq b - 1 - \left\lfloor \frac{\Delta_i - 1}{a} \right\rfloor\right\}.}$$

Proof. Let $\lambda_0 = \rho_{i+1}$, and let λ_* be the first nonmultiple of a at or after λ_0 :

$$\lambda_* = \begin{cases} \lambda_0, & a \nmid \lambda_0, \\ \lambda_0 + 1, & a \mid \lambda_0. \end{cases}$$

This target still lies below $2C$: the largest multiple of a below $2C$ is $2C - a$. At every nondivisible transition target, Corollary 7.4 gives

$$\nu_S(\lambda) = 2\#\{t \in T : at < \lambda - C\},$$

which is nondecreasing as λ increases.

Consider instead a divisible transition target $\lambda = az$. In this case, $b \leq z < 2b$, and Corollary 7.4 gives $\nu_S(az) = \nu_T(z)$. The following target $az + 1$ is nondivisible, with

$$\nu_S(az + 1) = 2\#\{t \in T : t \leq z - b\}.$$

For every $t \in T$ with $t \leq z - b$, one has

$$z - t \geq b \geq c(T),$$

so $z - t \in T$. Moreover, $z < 2b$ implies $t \leq z - b < z/2$, so the two ordered decompositions

$$z = t + (z - t), \quad z = (z - t) + t$$

are distinct. Distinct choices of t give distinct ordered pairs, and thus

$$\nu_T(z) \geq 2\#\{t \in T : t \leq z - b\} = \nu_S(az + 1).$$

No divisible transition target therefore lowers the suffix minimum below the immediately following nondivisible target.

Targets beyond the transition range cannot lower it either. Since the largest gap of S is $C - 1$, one has $R_S(\lambda) = 0$ for $\lambda \geq 2C - 1$. Consequently,

$$\nu_S(2C) = 2C + 1 - 2g(S) = 2(b - g(T)) + 1 = \nu_S(2C - 1) + 1.$$

Thereafter $\nu_S(\lambda) = \lambda + 1 - 2g(S)$ is increasing. It follows that

$$\boxed{\delta_i = \nu_S(\lambda_*)}. \tag{19}$$

In the two cases $a \nmid \lambda_0$ and $a \mid \lambda_0$, respectively, a direct division with remainder gives

$$\left\lfloor \frac{\lambda_* - C - 1}{a} \right\rfloor = b - 1 - \left\lfloor \frac{\Delta_i - 1}{a} \right\rfloor = m_i.$$

The strict inequality in the nondivisible branch of Corollary 7.4 therefore becomes $t \leq m_i$, proving the formula. This includes $\lambda_0 = C$, where $m_i = 0$ and $\delta_i = 2$, and $\lambda_0 = 2C - 1$, where $m_i = b - 1$. The argument also applies unchanged when $a = 2$. $\square$

Let

$$L_i = i - g(S) + 2 + \varepsilon_a(\Delta_i) \tag{20}$$

denote the AP-layer lower bound from Corollary 7.6.

Corollary 7.8 (AP-layer decomposition of the order bound). *For every transition index,*

$$\boxed{L_i = 2(m_i + 1 - g(T))}$$

and

$$\boxed{\delta_i = L_i + 2\#\{h \in G(T) : h > m_i\}}. \quad (21)$$

Equivalently,

$$\boxed{\delta_i = i - g(S) + 2 + \varepsilon_a(\Delta_i) + 2\#\left\{h \in G(T) : h \geq b - \left\lfloor \frac{\Delta_i - 1}{a} \right\rfloor\right\}}.$$

Proof. In both residue cases, the definition of k_i gives the unified identity

$$\varepsilon_a(\Delta_i) = \Delta_i - 1 - 2k_i.$$

Using $\rho_{i+1} = i + 1 + g(S)$, $\rho_{i+1} = 2C - \Delta_i$, $C = ab$, and $g(S) = (a - 1)b + g(T)$, we obtain

$$\begin{aligned} L_i &= \rho_{i+1} + 1 - 2g(S) + \varepsilon_a(\Delta_i) \\ &= 2b - 2g(T) - 2k_i = 2(m_i + 1 - g(T)). \end{aligned}$$

On the other hand,

$$\#(T \cap [0, m_i]) = m_i + 1 - \#(G(T) \cap [0, m_i]).$$

Substituting this in Theorem 7.7 and subtracting the displayed formula for L_i gives (21). Finally, $h > m_i$ is equivalent to $h \geq b - \lfloor (\Delta_i - 1)/a \rfloor$. $\square$

The exact Feng–Rao value consequently decomposes into a contribution determined solely by the new parallel gap-AP layer and a correction determined by the predecessor gap tail:

$$\boxed{\delta_i - L_i = 2\Theta_T(m_i)}, \quad \Theta_T(m) = \#\{h \in G(T) : h > m\}.$$

Thus the predecessor tail measures precisely the failure of the universal parallel-AP lower bound to be exact, and its vanishing determines the sharpness window.

Theorem 7.9 (AP-layer sharpness criterion). *For $C \leq \rho_{i+1} < 2C$, equality holds in Corollary 7.6,*

$$\delta_i = i - g(S) + 2 + \varepsilon_a(\Delta_i),$$

if and only if

$$\boxed{\Delta_i \leq C - ac(T) + a}.$$

Equivalently, the lower bound contributed by the new parallel gap-AP layer equals the full order bound precisely on a terminal subinterval of the conductor transition region. This is an AP-layer sharpness statement, not a claim that stabilization of the classical order bound was previously unknown.

Proof. By Corollary 7.8, sharpness is equivalent to the absence of predecessor gaps above m_i . With the usual convention $F(\mathbb{N}_0) = -1$, the Frobenius number is $F(T) = c(T) - 1$, so this is equivalent to

$$m_i \geq c(T) - 1.$$

Using (18), this becomes

$$\left\lfloor \frac{\Delta_i - 1}{a} \right\rfloor \leq b - c(T).$$

Since all quantities are integral, the last inequality is equivalent to

$$\Delta_i \leq a(b - c(T) + 1) = C - ac(T) + a.$$

□

For the Garcia–Stichtenoth tower, define

$$\boxed{H_n = c_n - qc_{n-1} + q.} \quad (22)$$

Theorem 7.9 says that the AP layer is sharp exactly when $\Delta_i \leq H_n$, or equivalently

$$\boxed{2c_n - H_n \leq \rho_{i+1} < 2c_n.}$$

Corollary 7.10 (Parity of the sharpness window). *For the Garcia–Stichtenoth semigroups,*

$$\boxed{H_n = \begin{cases} q, & n \text{ odd}, \\ q^{n/2}(q-1) + q, & n \text{ even}. \end{cases}}$$

It follows that the AP-layer bound is exact precisely on

$$2c_n - H_n \leq \rho_{i+1} < 2c_n,$$

intersected with the transition range. For $n = 2$, one has $H_2 = q^2 > c_2$, so the entire transition interval is sharp.

Proof. If $n = 2r + 1$, then $c_{2r+1} = qc_{2r}$, and (22) gives $H_{2r+1} = q$. If $n = 2r$, then

$$c_{2r} - qc_{2r-1} = q^r(q-1),$$

and hence $H_{2r} = q^r(q-1) + q$. □

At odd levels, $c_n = qc_{n-1}$: the new cutoff introduces no interval between the scaled predecessor conductor and the new conductor. Predecessor gaps therefore remain relevant almost throughout the transition range, and the pure parallel-AP contribution becomes exact only in the final q positions. At even levels,

$$c_n - qc_{n-1} = q^{n/2}(q-1)$$

is large. This produces a substantially longer terminal interval in which the predecessor gap tail has disappeared and the universal parallel AP layer alone determines the exact order bound. Theorem 7.9 resolves the AP-layer sharpness question for one inductive step, while Corollary 7.10 gives its parity-dependent tower form.

The predecessor correction can itself be resolved into layers inherited from earlier tower levels. This resolution connects the exact profile theory of Section 5 directly with (21). Let

$$G_n = G(\Lambda_n), \quad U_n = \{1 \leq x < c_n : q \nmid x\}.$$

The tower recursion gives the disjoint decomposition

$$G_n = U_n \sqcup qG_{n-1}. \quad (23)$$

Proposition 7.11 (Tower gap blocks in AP form). *For every $n \geq 2$,*

$$G_n = \bigsqcup_{k=2}^n q^{n-k} U_k. \quad (24)$$

If $b_k = c_k/q$, then equivalently

$$G_n = \bigsqcup_{k=2}^n \bigsqcup_{r=1}^{q-1} \{q^{n-k}r + \ell q^{n-k+1} : 0 \leq \ell < b_k\}. \quad (25)$$

Consequently, the layer inherited from level k consists of $q - 1$ arithmetic progressions of common difference q^{n-k+1} and length b_k .

Proof. The case $n = 2$ is (23), since G_1 is empty. If the result holds at level $n - 1$, then

$$G_n = U_n \sqcup qG_{n-1} = U_n \sqcup \bigsqcup_{k=2}^{n-1} q^{n-k} U_k,$$

which is (24). Disjointness is inherited at each step; equivalently, elements of $q^{n-k}U_k$ are divisible by q^{n-k} but not by q^{n-k+1} . Finally,

$$U_k = \bigsqcup_{r=1}^{q-1} \{r, r + q, \dots, r + (b_k - 1)q\},$$

and scaling by q^{n-k} gives (25). $\square$

Nonrecursive descriptions and enumerations of the Garcia–Stichtenoth semigroups are already known, notably in the work used by Bras-Amorós and O’Sullivan [2]. The present formulation reads these gap blocks as arithmetic progressions, identifies their lengths with the profile, and uses them to decompose the Feng–Rao discrepancy.

Proposition 7.12 (Profile length of an ancestral layer). *If $q = p^s$, then for every $k \geq 2$,*

$$\mathcal{A}_k(s) = \frac{c_k}{q} = b_k.$$

Accordingly, the length of each AP in the ancestral k -th layer is exactly the $v_p(d) = s$ component of the AP profile of Λ_k .

Proof. Theorem 5.6 at $t = s$ gives

$$\mathcal{A}_k(s) = \max \left\{ \left\lceil \frac{c_k - 1}{q} \right\rceil, \mathcal{A}_{k-1}(0) \right\} = \max \{c_k/q, q^{k-2} - 1\}.$$

Let $r = \lfloor (k+1)/2 \rfloor$. Since $k - 2 \geq r - 1$,

$$\frac{c_k}{q} - (q^{k-2} - 1) = q^{k-2}(q - 1) - q^{r-1} + 1 \geq 1.$$

The conductor branch is therefore strictly larger. $\square$

Define the tail-gap counting function

$$\Theta_n(m) = \#\{h \in G_n : h > m\},$$

with $\Theta_n(m) = 0$ for $m \geq c_n - 1$, and use the notation $[x]_+ = \max\{x, 0\}$.

Theorem 7.13 (Tail-gap recursion). *For $n \geq 2$ and $0 \leq m < c_n$,*

$$\Theta_n(m) = (q-1)b_n - m + \left\lfloor \frac{m}{q} \right\rfloor + \Theta_{n-1}\left(\left\lfloor \frac{m}{q} \right\rfloor\right). \quad (26)$$

Equivalently, using Proposition 7.12, the uniform formula valid for every $m \geq 0$ is

$$\Theta_n(m) = \left[(q-1)\mathcal{A}_n(s) - m + \left\lfloor \frac{m}{q} \right\rfloor \right]_+ + \Theta_{n-1}\left(\left\lfloor \frac{m}{q} \right\rfloor\right). \quad (27)$$

Proof. Among the $c_n - 1 - m$ integers in $\{m+1, \dots, c_n - 1\}$, exactly

$$\frac{c_n}{q} - 1 - \left\lfloor \frac{m}{q} \right\rfloor$$

are divisible by q . Subtracting gives

$$\#\{u \in U_n : u > m\} = (q-1)b_n - m + \left\lfloor \frac{m}{q} \right\rfloor.$$

The scaled predecessor gaps contribute

$$\#\{h \in G_{n-1} : qh > m\} = \Theta_{n-1}\left(\left\lfloor \frac{m}{q} \right\rfloor\right),$$

which proves (26). At $m = c_n - 1$ the new-layer expression is zero; thereafter it is nonpositive. Moreover, $\lfloor m/q \rfloor \geq b_n - 1 \geq c_{n-1} - 1$, so the predecessor tail also vanishes. This proves the clipped formula for all $m \geq 0$. $\square$

Corollary 7.14 (Ancestral AP reconstruction of the tail). *For every $n \geq 2$ and $m \geq 0$,*

$$\Theta_n(m) = \sum_{k=2}^n \left[(q-1)\mathcal{A}_k(s) - \left\lfloor \frac{m}{q^{n-k}} \right\rfloor + \left\lfloor \frac{m}{q^{n-k+1}} \right\rfloor \right]_+. \quad (28)$$

Proof. Iterate (27). At ancestral level k , the threshold is $\lfloor m/q^{n-k} \rfloor$, and division once more by q gives $\lfloor m/q^{n-k+1} \rfloor$. The recursion terminates at $G_1 = \emptyset$. $\square$

The k -th summand in (28) counts the elements above m in the terminal pieces of the $q-1$ scaled APs inherited from level k . Thus the formula expresses the entire tail-gap count as the sum of the terminal ancestral AP-layer contributions.

Specializing Corollary 7.8 to $S = \Lambda_n$ and $T = \Lambda_{n-1}$ gives

$$L_i = i - g_n + 2 + \varepsilon_q(\Delta_i), \quad \delta_i - L_i = 2\Theta_{n-1}(m_i).$$

Theorem 7.15 (AP-layer decomposition of the tower discrepancy). *For every transition index at level n ,*

$$\delta_i = L_i + 2 \sum_{k=2}^{n-1} \left[(q-1)\mathcal{A}_k(s) - \left\lfloor \frac{m_i}{q^{n-1-k}} \right\rfloor + \left\lfloor \frac{m_i}{q^{n-k}} \right\rfloor \right]_+. \quad (29)$$

For $n = 2$ the sum is empty. The predecessor-tail discrepancy is therefore exactly reconstructible from the ancestral profile values $\mathcal{A}_2(s), \dots, \mathcal{A}_{n-1}(s)$.

Proof. Substitute (28), with $n - 1$ in place of n and m_i in place of m , into $\delta_i - L_i = 2\Theta_{n-1}(m_i)$. $\square$

In this form, the exact order bound is the sum of the current parallel-AP contribution and the ancestral corrections. In particular, $L_i = \delta_i$ if and only if every ancestral correction summand vanishes. The terminal sharpness window is therefore the final region in which all inherited AP layers have disappeared. Before this window is reached, different ancestral summands vanish as their thresholds are crossed; no ordering of those disappearances is asserted here.

Corollary 7.16 (AP reformulation of the tower genus formula). *For every $n \geq 2$,*

$$g_n = (q - 1) \sum_{k=2}^n \mathcal{A}_k(s).$$

Proof. Taking cardinalities in (24) gives

$$g_n = \sum_{k=2}^n |U_k|.$$

Each U_k consists of $q - 1$ APs of length $b_k = \mathcal{A}_k(s)$, so $|U_k| = (q - 1)\mathcal{A}_k(s)$. $\square$

The genus formula itself is standard; here it is reformulated as cumulative gap-AP-layer mass. This complements Corollary 6.2, although it does not by itself explain the asymptotic characteristic slope geometrically.

The reconstruction above uses the ancestral collection $\{\mathcal{A}_k(s)\}_{2 \leq k < n}$. A q -depth refinement introduced below reads these ancestral gap-AP lengths from a single tower level. Whether the unrefined spectrum contains the same information remains open; see Problem 9.1.

The formulas in Sections 5–8 were independently checked by direct enumeration at several low levels of the Garcia–Stichtenoth tower, including $q = 2, 3, 4, 5, 8, 9$, and on several nonconstant inductive examples. The checks cover the AP profile, gap-pair recursion, AP-layer discrepancy, sharpness criterion, ancestral tail reconstruction, and q -depth identity. These calculations are supplementary and do not enter the proofs. Verification code is available from the author.

8 Inductive numerical semigroups

The block structure of inductive numerical semigroups and their Feng–Rao and Apéry invariants are well established [3, 5]. We use this structure on the complementary gap side to determine how its standard parameters are detected by the gap-AP invariants of Section 2.

Consider the standard inductive construction

$$S_1 = \mathbb{N}_0, \quad S_k = a_k S_{k-1} \cup [C_k, \infty) \quad (k \geq 2), \quad (30)$$

where

$$a_k \geq 2, \quad C_k = a_k b_k, \quad b_k \geq c(S_{k-1}).$$

Denote the gap set of S_k by G_k and define

$$U_k = \{1 \leq x < C_k : a_k \nmid x\}.$$

The fixed-difference proof in Section 5 does not require a prime-power scale, so it applies unchanged to each step of this construction.

Proposition 8.1 (Divisible-difference AP recursion). *Let T be a numerical semigroup with conductor $c(T)$, let $a \geq 2$, and let*

$$S = aT \cup [C, \infty)$$

be a numerical semigroup with $C \geq ac(T)$. If $a \mid d$, then

$$\boxed{\text{AP}_d(S) = \max \left\{ \left\lceil \frac{C-1}{d} \right\rceil, \text{AP}_{d/a}(T) \right\}}.$$

Proof. If $C = 0$, then $T = S = \mathbb{N}_0$, and both terms in the formula vanish. Assume $C > 0$. The progression has constant residue modulo a . In a nonzero residue, all terms below C are gaps and the best start is 1. In residue zero, division by a identifies gap progressions with those of T ; the condition $C \geq ac(T)$ ensures that scaling a gap of T stays below C . This is the same proof as Proposition 5.5. $\square$

The standard inductive-semigroup decomposition [5] has the following complementary gap-side form. For every $k \geq 2$,

$$\boxed{G_k = U_k \sqcup a_k G_{k-1}}. \quad (31)$$

For $2 \leq k \leq n$, define

$$A_{k,n} = \prod_{j=k+1}^n a_j, \quad A_{n,n} = 1. \quad (32)$$

It follows that

$$\boxed{G_n = \bigsqcup_{k=2}^n A_{k,n} U_k}, \quad (33)$$

where

$$A_{k,n} U_k = \bigsqcup_{r=1}^{a_k-1} \{A_{k,n} r + \ell a_k A_{k,n} : 0 \leq \ell < b_k\}. \quad (34)$$

Consequently, the ancestral k -block consists of $a_k - 1$ parallel gap APs, each of length b_k and common difference $a_k A_{k,n} = \prod_{j=k}^n a_j$.

To verify the gap-side form directly, observe that below C_k a nonmultiple of a_k lies in U_k , while a multiple $a_k h$ is a gap precisely when $h \in G_{k-1}$. The inequality $C_k \geq a_k c(S_{k-1})$ ensures that every scaled predecessor gap lies below C_k , which verifies (31) directly on the gap side. Iteration gives (33). Moreover,

$$U_k = \bigsqcup_{r=1}^{a_k-1} \{r, r + a_k, \dots, r + (b_k - 1)a_k\},$$

so scaling by $A_{k,n}$ gives (34).

For the present argument, these standard blocks are viewed as arithmetic-progressive gap layers whose lengths can be compared with the invariants of Section 2.

Proposition 8.2 (Inductive layer length as a gap-AP invariant). *For every $k \geq 2$,*

$$\boxed{\text{AP}_{a_k}(S_k) = b_k}. \quad (35)$$

Consequently, the standard construction parameter b_k is exactly the fixed-difference gap-AP length at the stage where the layer is created.

Proof. Proposition 8.1, applied with $d = a_k$, gives

$$\text{AP}_{a_k}(S_k) = \max \left\{ \left\lceil \frac{C_k - 1}{a_k} \right\rceil, \text{AP}_1(S_{k-1}) \right\} = \max\{b_k, \text{AP}_1(S_{k-1})\}.$$

If $S_{k-1} = \mathbb{N}_0$, then $\text{AP}_1(S_{k-1}) = 0 \leq b_k$. Otherwise, Lemma 2.3 and $m(S_{k-1}) \leq c(S_{k-1}) \leq b_k$ give $\text{AP}_1(S_{k-1}) \leq b_k - 1$. $\square$

Although b_k is standard construction data, its explicit identification with a maximal fixed-difference gap-AP invariant does not appear in the inductive-semigroup literature known to the author. The proposition concerns this AP interpretation, rather than the parameter or block structure itself.

For the Garcia–Stichtenoth family, $a_k = q = p^s$, and this specializes to

$$\text{AP}_q(\Lambda_k) = \mathcal{A}_k(s) = c_k/q.$$

For $m \geq 0$, define the tail function by

$$\Theta_k(m) = \#\{h \in G_k : h > m\}, \quad [x]_+ = \max\{x, 0\}.$$

Theorem 8.3 (AP-layer reconstruction of the gap tail). *For every $k \geq 2$ and $m \geq 0$,*

$$\Theta_k(m) = \left[(a_k - 1) \text{AP}_{a_k}(S_k) - m + \left\lfloor \frac{m}{a_k} \right\rfloor \right]_+ + \Theta_{k-1} \left(\left\lfloor \frac{m}{a_k} \right\rfloor \right). \quad (36)$$

Iterating this identity gives

$$\Theta_n(m) = \sum_{k=2}^n \left[(a_k - 1) \text{AP}_{a_k}(S_k) - \left\lfloor \frac{m}{A_{k,n}} \right\rfloor + \left\lfloor \frac{m}{a_k A_{k,n}} \right\rfloor \right]_+. \quad (37)$$

Proof. If $C_k = 0$, then $S_{k-1} = S_k = \mathbb{N}_0$, and the recursive identity is immediate at this stage. Assume $C_k > 0$. For $0 \leq m < C_k$, the interval $\{m+1, \dots, C_k-1\}$ contains $C_k - 1 - m$ integers, of which

$$b_k - 1 - \left\lfloor \frac{m}{a_k} \right\rfloor$$

are divisible by a_k . The contribution of U_k is therefore $(a_k - 1)b_k - m + \lfloor m/a_k \rfloor$. The scaled predecessor gaps contribute $\Theta_{k-1}(\lfloor m/a_k \rfloor)$. At $m = C_k - 1$ the new-layer expression vanishes and is nonpositive thereafter; the predecessor tail also vanishes because $\lfloor m/a_k \rfloor \geq b_k - 1 \geq c(S_{k-1}) - 1$. Proposition 8.2 therefore gives (36) for all $m \geq 0$.

Iterating the clipped formula, the threshold at ancestral level k is $\lfloor m/A_{k,n} \rfloor$, and one more division gives $\lfloor m/(a_k A_{k,n}) \rfloor$. This proves (37). $\square$

The k -th summand in (37) is exactly the terminal part above m of the ancestral k -th gap-AP layer. The formula is therefore an AP-layer expression for the gap tail, rather than a new structural classification of inductive semigroups.

At the final inductive step, use the notation of Section 7:

$$L_i = i - g(S_n) + 2 + \varepsilon_{a_n}(\Delta_i), \quad m_i = b_n - 1 - \left\lfloor \frac{\Delta_i - 1}{a_n} \right\rfloor.$$

We write $S_n = \{\rho_0 = 0 < \rho_1 < \dots\}$.

Theorem 8.4 (AP-layer decomposition of the Feng–Rao discrepancy). *If $C_n \leq \rho_{i+1} < 2C_n$, then*

$$\delta_i = L_i + 2 \sum_{k=2}^{n-1} \left[(a_k - 1) \text{AP}_{a_k}(S_k) - \left\lfloor \frac{m_i}{A_{k,n-1}} \right\rfloor + \left\lfloor \frac{m_i}{a_k A_{k,n-1}} \right\rfloor \right]_+. \quad (38)$$

For $n = 2$ the sum is empty.

Proof. Corollary 7.8 gives $\delta_i - L_i = 2\Theta_{n-1}(m_i)$. Substitute (37) with $n - 1$ in place of n . $\square$

The full order bound is known in the Arf and inductive literature [3, 5]. The content of Theorem 8.4 is its AP-layer decomposition: the discrepancy from the current-layer estimate is reconstructed exactly as a sum of ancestral gap-AP contributions.

The same identification gives an AP interpretation of the standard genus formula. Namely,

$$g(S_n) = \sum_{k=2}^n (a_k - 1)b_k$$

is standard for inductive semigroups; see, for example, [5]. Combining it with Proposition 8.2 gives the AP interpretation

$$g(S_n) = \sum_{k=2}^n (a_k - 1) \text{AP}_{a_k}(S_k). \quad (39)$$

The displayed equality records the standard genus formula in terms of fixed-difference gap APs.

For the constant scale of the Garcia–Stichtenoth tower, one can also retain the starting-point depth relative to the distinguished scale q . Standard structural data, such as consecutive-nongap differences, already reflect the inductive ancestry; the refinement below records specifically the ancestral *gap-AP lengths* visible from one level.

Definition 8.5 (q -depth AP profile). For $n \geq 2$ and $0 \leq r \leq n - 2$, define

$$\mathcal{D}_n(r) = \max \left\{ \ell : \begin{array}{l} x, x + q^{r+1}, \dots, x + (\ell - 1)q^{r+1} \in G_n, \\ q^r \mid x, \quad q^{r+1} \nmid x \end{array} \right\}.$$

Proposition 8.6 (Recovery of ancestral gap-AP lengths). For $0 \leq r \leq n - 2$,

$$\mathcal{D}_n(r) = \frac{c_{n-r}}{q} = \mathcal{A}_{n-r}(s). \quad (40)$$

Proof. Divide a progression counted by $\mathcal{D}_n(r)$ by q exactly r times, using

$$qh \in G(\Lambda_j) \iff h \in G(\Lambda_{j-1}).$$

The result is a q -step gap AP in Λ_{n-r} whose start is not divisible by q . It remains in a fixed nonzero residue class and hence is a gap exactly until it reaches c_{n-r} . Its length is at most c_{n-r}/q , attained by the canonical APs in U_{n-r} . Scaling such a progression by q^r proves the converse. Proposition 7.12 gives the final equality. $\square$

The ancestral sequence of gap-AP lengths is thus readable from refined AP data of the single semigroup Λ_n . The tower discrepancy formula may be rewritten as

$$\delta_i = L_i + 2 \sum_{r=0}^{n-3} \left[(q-1) \mathcal{D}_{n-1}(r) - \left\lfloor \frac{m_i}{q^r} \right\rfloor + \left\lfloor \frac{m_i}{q^{r+1}} \right\rfloor \right]_+,$$

with the sum understood as empty for $n = 2$. Likewise,

$$g_n = (q-1) \sum_{r=0}^{n-2} \mathcal{D}_n(r).$$

These formulas recover the AP-layer data without reconstructing the entire semigroup.

9 Open problems

Problem 9.1 (Full AP spectrum and maximizers). Determine or efficiently encode the full spectrum $\Sigma(\Lambda_n)$, classify the pairs (x, d) that attain $\mathcal{A}_n(t)$ in each p -adic layer, and decide whether the ordinary spectrum already determines the q -depth profile $\mathcal{D}_n(r)$. The last question asks whether starting-point q -adic depth is genuinely additional data or is encoded by the values $\text{AP}_d(\Lambda_n)$ alone.

Problem 9.2 (Depth profiles for varying scales). For a general inductive sequence with nonconstant scales a_k , develop an appropriate start-depth or divisor-lattice refinement that recovers the ancestral gap-AP lengths $\text{AP}_{a_k}(S_k)$ from one level without explicit knowledge of the construction history.

Problem 9.3 (Geometry of the characteristic slope). Determine whether the limit

$$\lim_{n \rightarrow \infty} \frac{g_n}{\text{AP}(\Lambda_n)} = p$$

has a geometric explanation beyond the explicit semigroup recursion, and whether analogous characteristic-detection occurs in other towers.

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